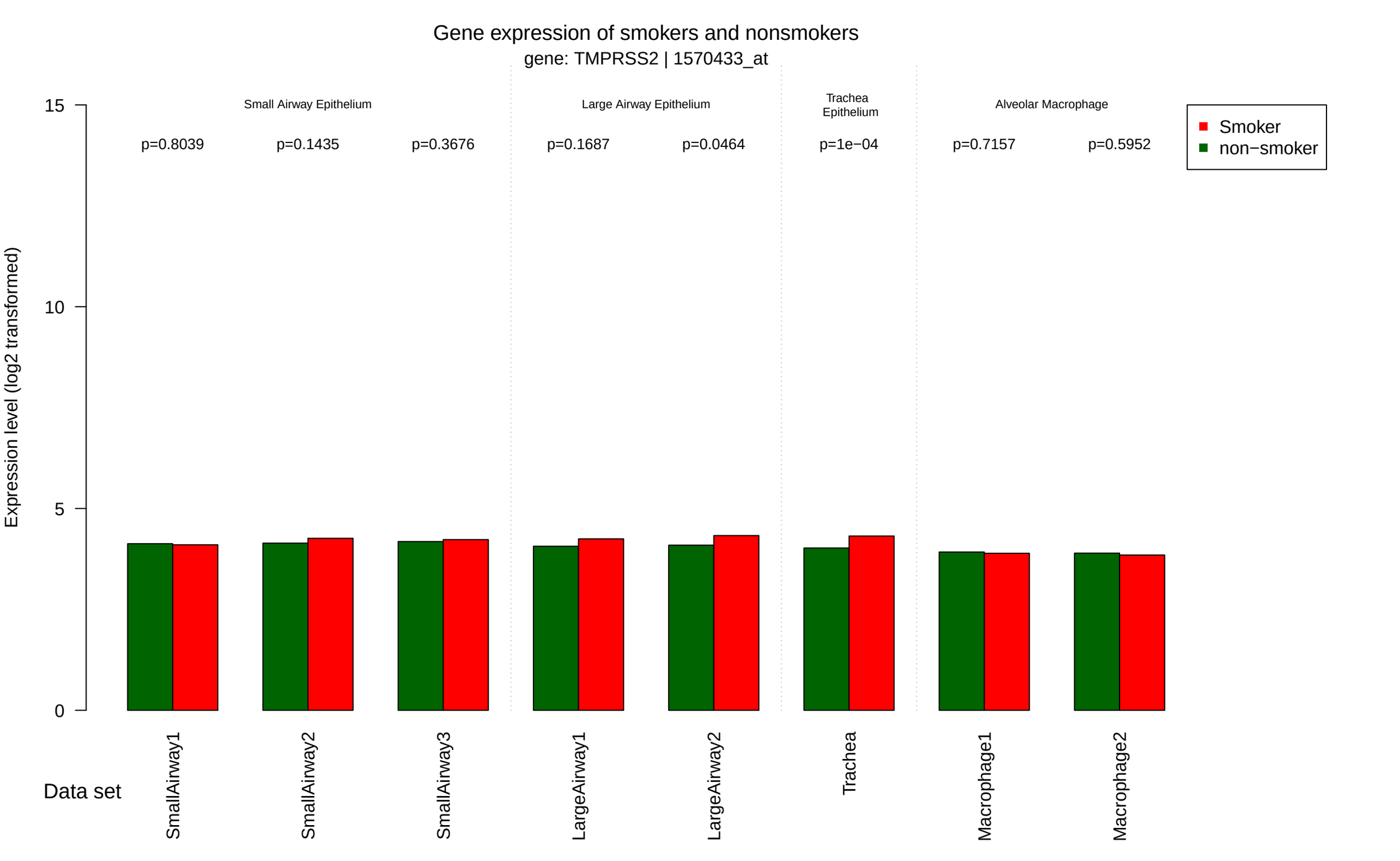
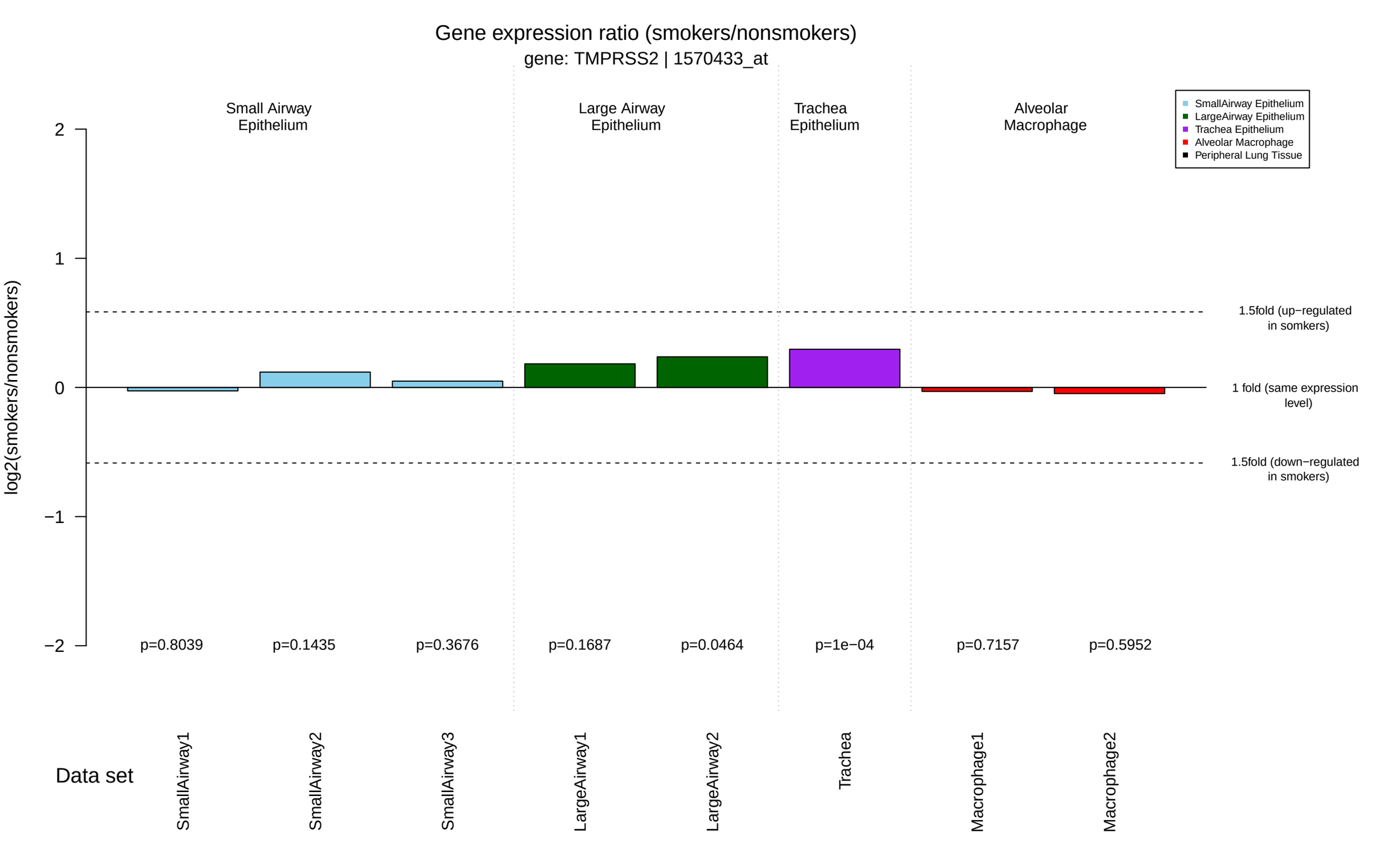
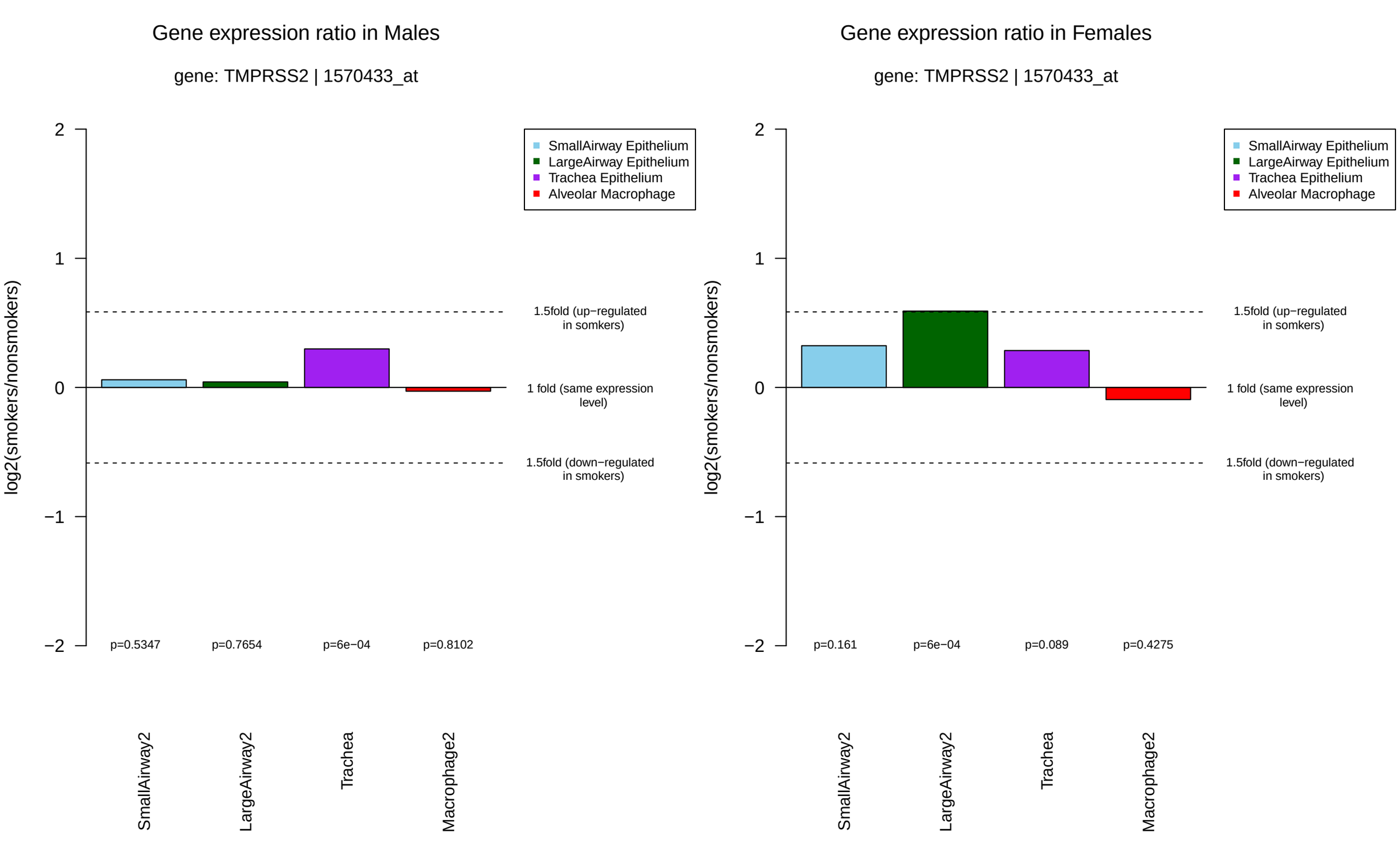
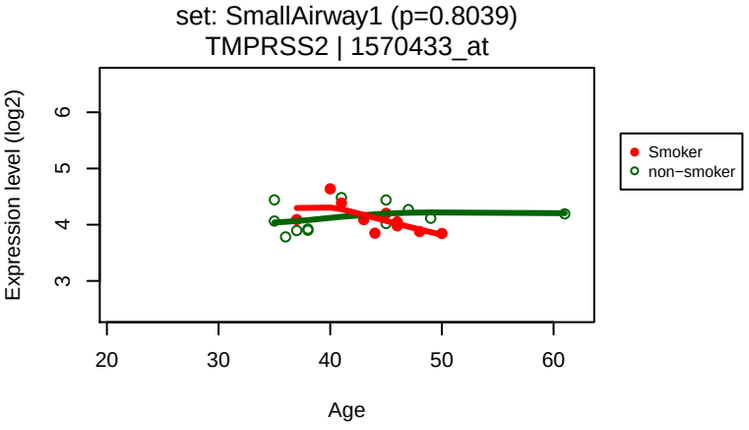
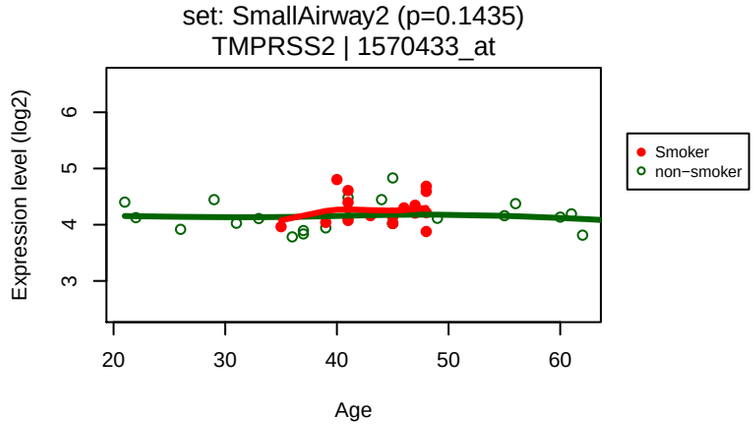
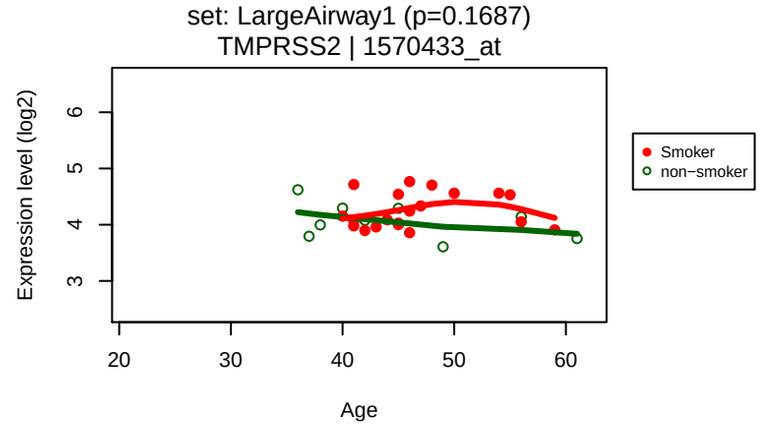

Gene expression of smokers and nonsmokers
gene: TMPRSS2 | 1570433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8039
p=0.1435
p=0.3676
p=0.1687
p=0.0464
p=1e−04
p=0.7157
p=0.5952
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

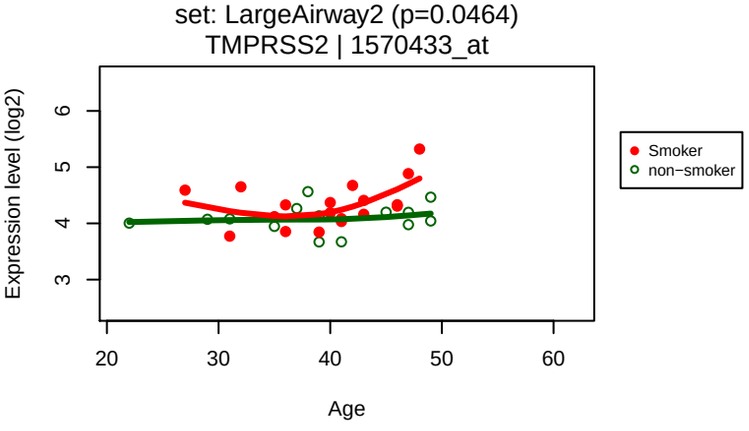

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS2 | 1570433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8039
p=0.1435
p=0.3676
p=0.1687
p=0.0464
p=1e−04
p=0.7157
p=0.5952
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

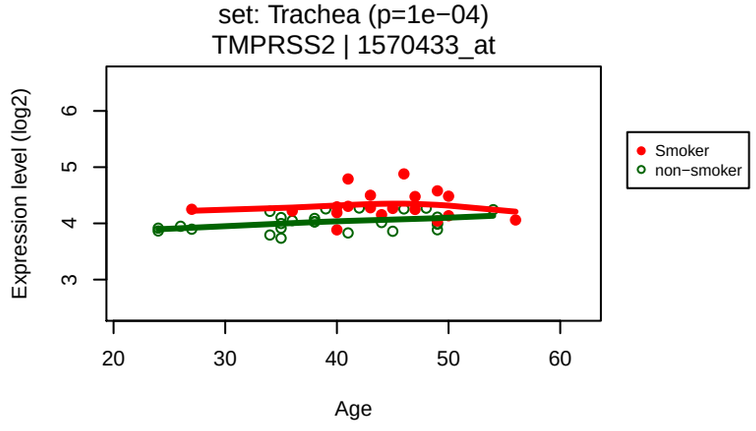

Gene expression ratio in Males
gene: TMPRSS2 | 1570433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5347
p=0.7654
p=6e−04
p=0.8102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS2 | 1570433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.161
p=6e−04
p=0.089
p=0.4275
SmallAirway2
LargeAirway2
Trachea
Macrophage2

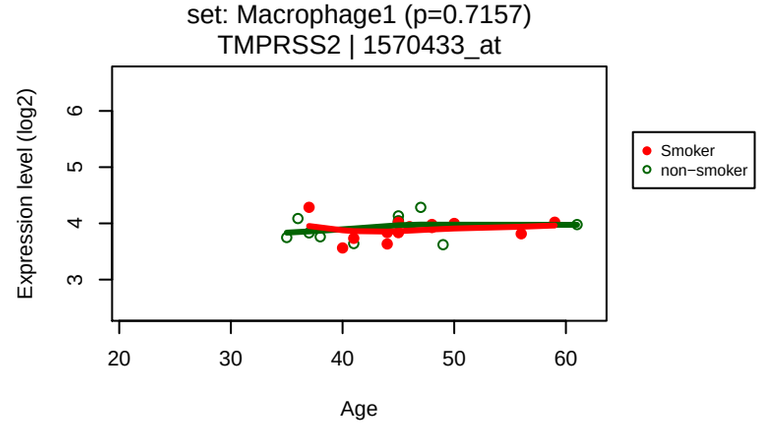

set: SmallAirway1 (p=0.8039)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker

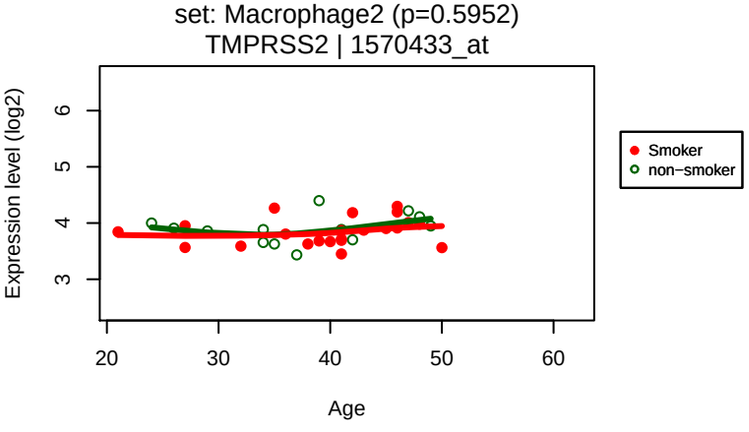

set: SmallAirway2 (p=0.1435)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker

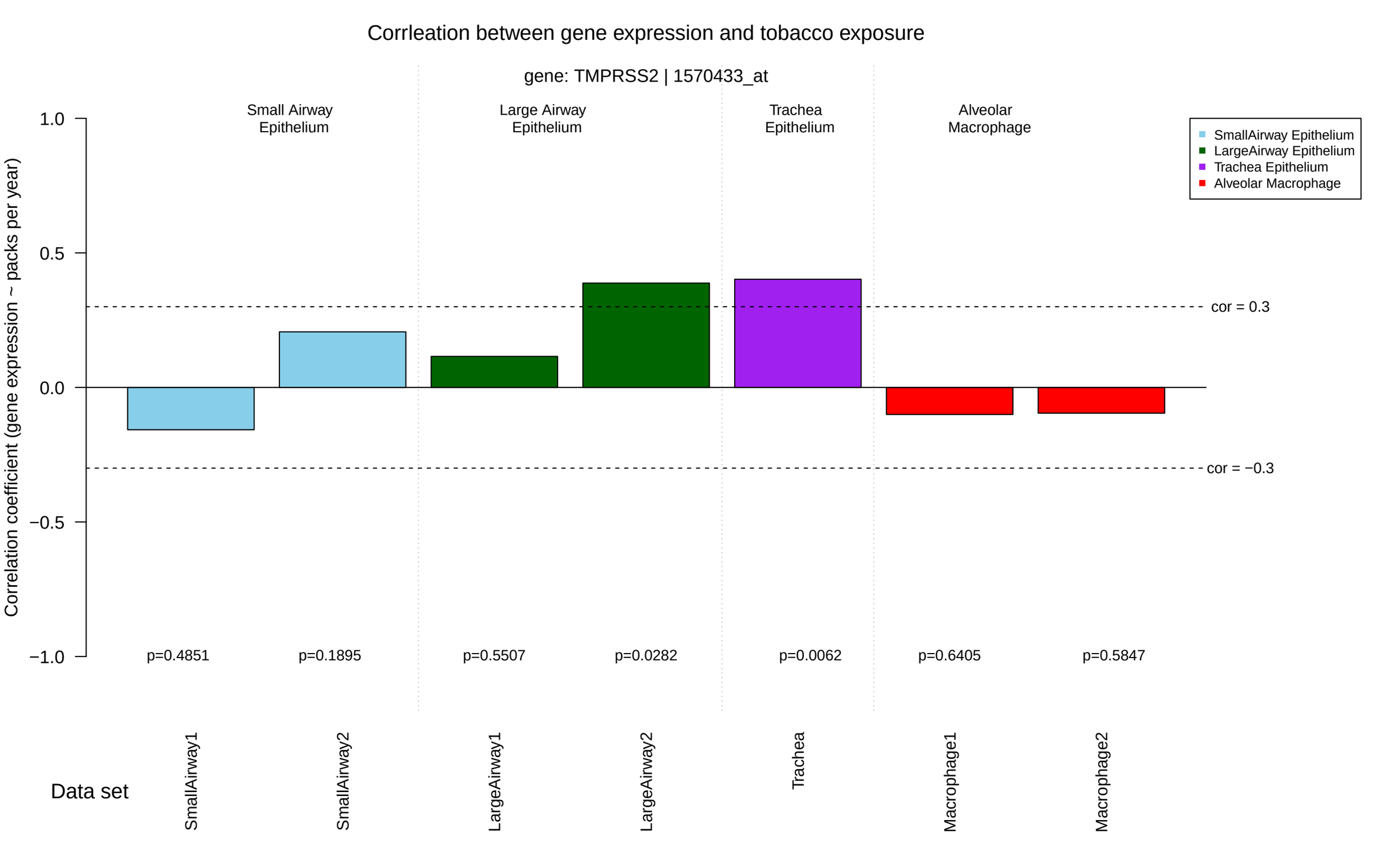

set: LargeAirway1 (p=0.1687)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0464)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7157)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5952)
TMPRSS2 | 1570433_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS2 | 1570433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4851
p=0.1895
p=0.5507
p=0.0282
p=0.0062
p=0.6405
p=0.5847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2